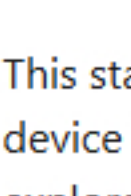
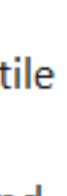
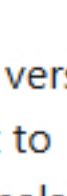




Raspberry Pi 4B: A Small, Battery Powered IT Tool / WiFi Extender / Media Device . . . & more.



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Part 1: Introduction

This started as a simple cybersecurity curiosity. I did not expect it to turn into such a versatile device with so many different types of fun packed into one little box. I first bought it to explore *Kali Linux*, which is a Linux distro filled with tools for cybersecurity professionals and hobbyists. Almost immediately after getting comfortable with *Linux*, I felt "at one with the machine." A lot of IT tasks I had done in the past started to make more sense, especially anything related to networking.

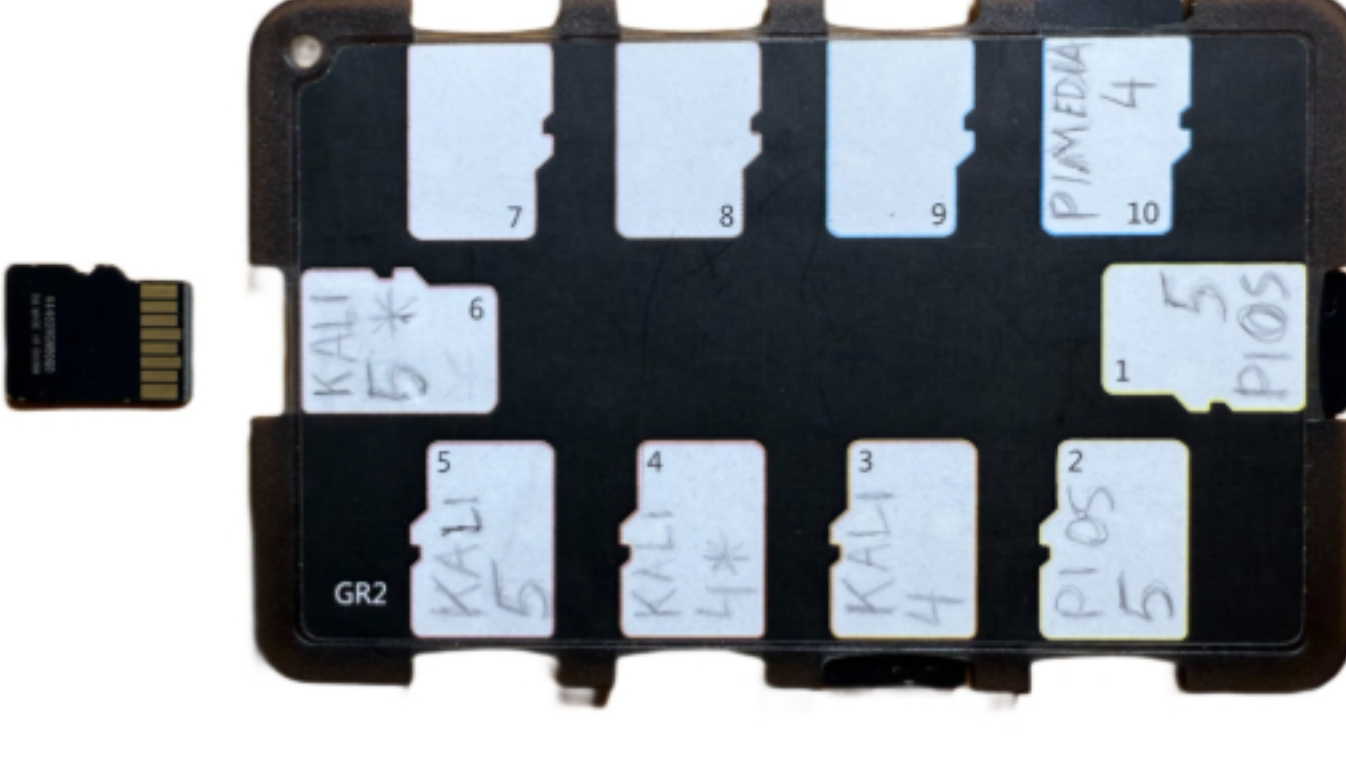
What you're seeing above is a *Raspberry Pi 4B* in a small case with a 4" touchscreen. It is powered by a battery bank that *Velcros* on and off. The setup includes a tiny wireless keyboard with a built-in mousepad. I also have a *USB* network adapter plugged in. With the adapter the battery lasts around 6 hours. Without it, the battery lasts about 9. The whole thing cost me around \$200.

At some point, it also became a useful battery powered *WiFi* extender and hotspot. I used it to stream movies outdoors with a projector. Later I realized it was the most efficient and convenient tool for certain IT jobs. And now, most of the time, it is not a terrible media device either.



4" LCD Raspberry Pi 4B with a Velcro mounted battery & mini keyboard. The writing pen is there for scale.

How does it work? A *Raspberry Pi* is a tiny *Linux* computer with a huge list of optional components that let you customize it for any application imaginable. For example, I once saw someone turn one into a rotating salt gun turret that detects mosquitoes with AI and shoots them.



SD Card Holder

The holder can store up to 10 SD cards. I am only using 7 slots so far. These are split between this device and another *Raspberry Pi 5* I built. About half of the cards are duplicate twins for quick backup recovery.

If you call the *Raspberry Pi 4B* in this article a "laptop" since it is portable, then my *Raspberry Pi 5* build would be the desktop version. You need to sit at a desk, plug it into an outlet and connect a monitor. I am saving that one for a future blog, because this is getting long very quickly.

Before we continue, if you want to learn from my mistakes and save a dozen hours of spinning in circles, this part may help you. If you have never used *Linux*, start with the fundamentals. Get comfortable with the terminal and basic commands before you jump into anything "cool" like hunting for weak spots in your network. That also means starting with the *Pi OS* operating system instead of *Kali Linux*.

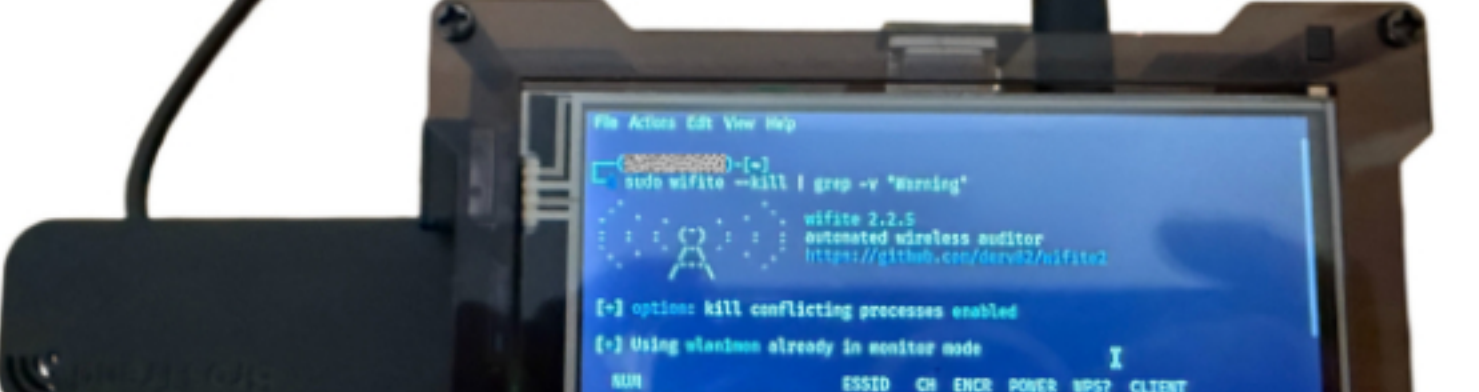
To help you get started, here is a basic *Linux* commands cheat sheet: [click here to view it on GitHub](#).

Another good tip is to stay away from "shady Chinese images". An image is basically a virtual hard drive that can be cloned to an SD card or another physical device. Some of these images are tempting because they come preconfigured, but it is better to take a little extra time and set everything up yourself from a verified source. Use *Raspberry Pi Imager*, which makes installation a breeze and has official releases ready for you to pick from.

Also, do not start learning on such a tiny device. While you are exploring everything, plug the *Pi* into a full size monitor, keyboard and mouse. Lastly, keep lots of backups, especially once you reach a configuration that you enjoy using.

Part 2: Small IT Tool

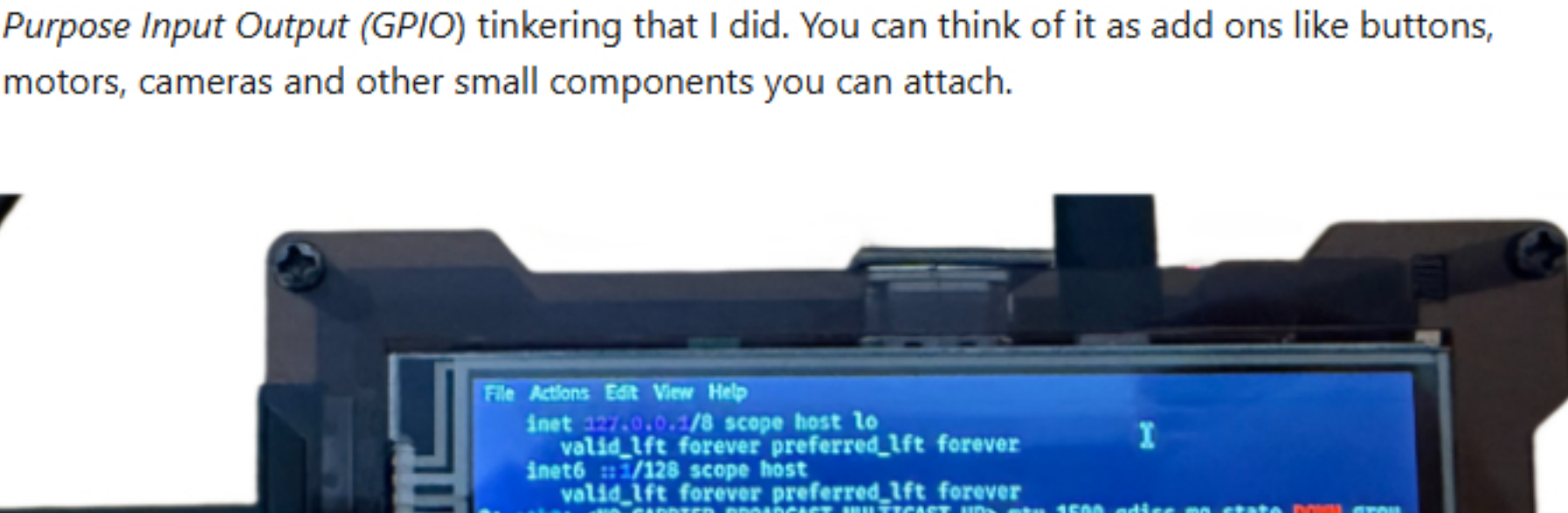
Although there is nothing this build does that a regular laptop could not, the size is what makes it such a good IT tool. It is much more convenient to walk around with something this small while testing a specific access point's *WiFi* strength or identifying ethernet ports in hard to reach locations. It is also a great way to demonstrate to an owner how a malicious actor with a tiny device can access a network if there is a vulnerability.



WiFiite

In the image above, you can see the *WiFiite* application. It automates the process of capturing a handshake, which can sometimes reveal your *WiFi* password. This requires an external *USB* network adapter and it gets a bit more involved. Do not underestimate a 13 year old with a lot of free time and access to *YouTube*....

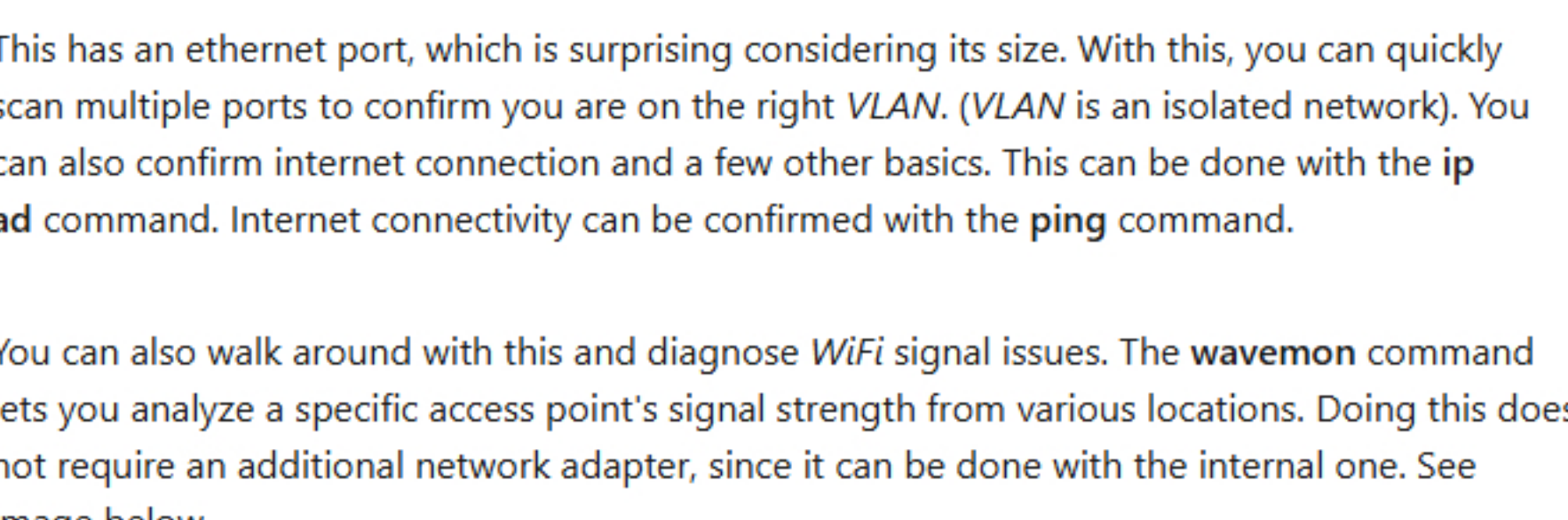
In a future blog about my other "desktop" *Raspberry Pi 5*, I will go over some of the cybersecurity experiments I tried. Once you find weak points, you can improve your setup by addressing them one by one. It is a back-and-forth process that needs occasional upkeep or at least a habit of following news about new exploits. There, I will also cover some fun *General Purpose Input Output (GPIO)* tinkering that I did. You can think of it as add ons like buttons, motors, cameras and other small components you can attach.



Network Adapter Information ("ip ad" command)

This has an ethernet port, which is surprising considering its size. With this, you can quickly scan multiple ports to confirm you are on the right *VLAN*. (*VLAN* is an isolated network). You can also confirm internet connection and a few other basics. This can be done with the *ip ad* command. Internet connectivity can be confirmed with the *ping* command.

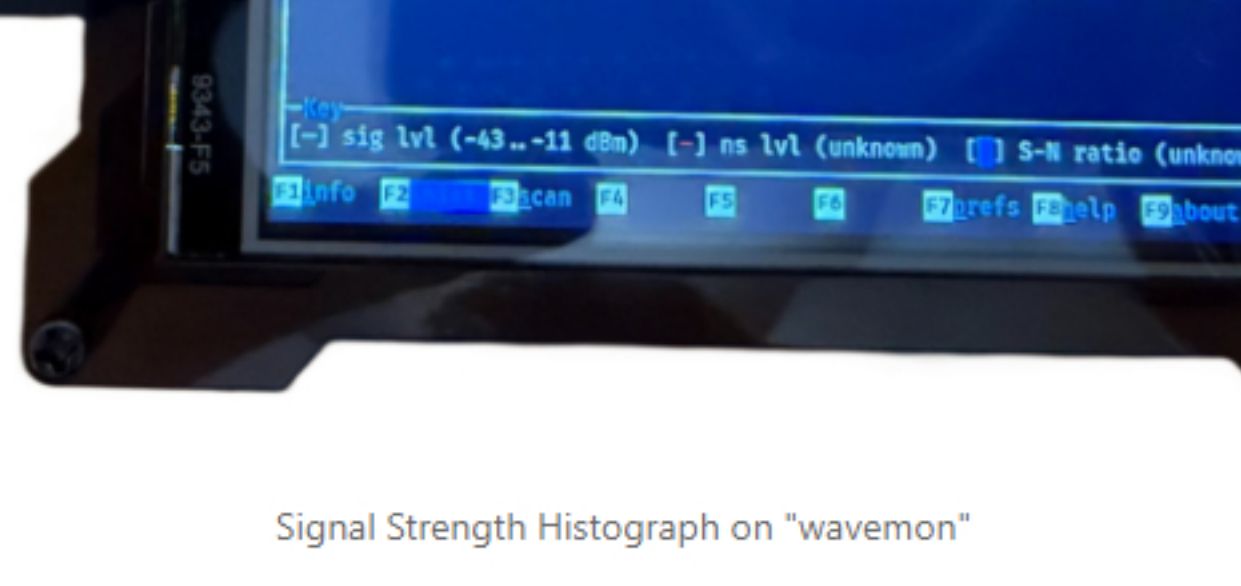
You can also walk around with this and diagnose *WiFi* signal issues. The *wavemon* command lets you analyze a specific access point's signal strength from various locations. Doing this does not require an additional network adapter, since it can be done with the internal one. See image below.



Signal Strength Histogram on "wavemon"

The one thing is not possible is detecting if an ethernet port has *Power Over Ethernet (POE)*. Even though they sell *POE* hats for the *Raspberry Pi*, those are only used to power the *Pi* itself. If someone knows a way to detect *POE* with a *Pi*, please let me know. Luckily, separate *POE* detection devices are pretty cheap. In contrast, a small specialty IT tool that handles all the tasks mentioned here would cost a lot more than this build, and would end up doing much less....

Part 3: Portable WiFi Extender

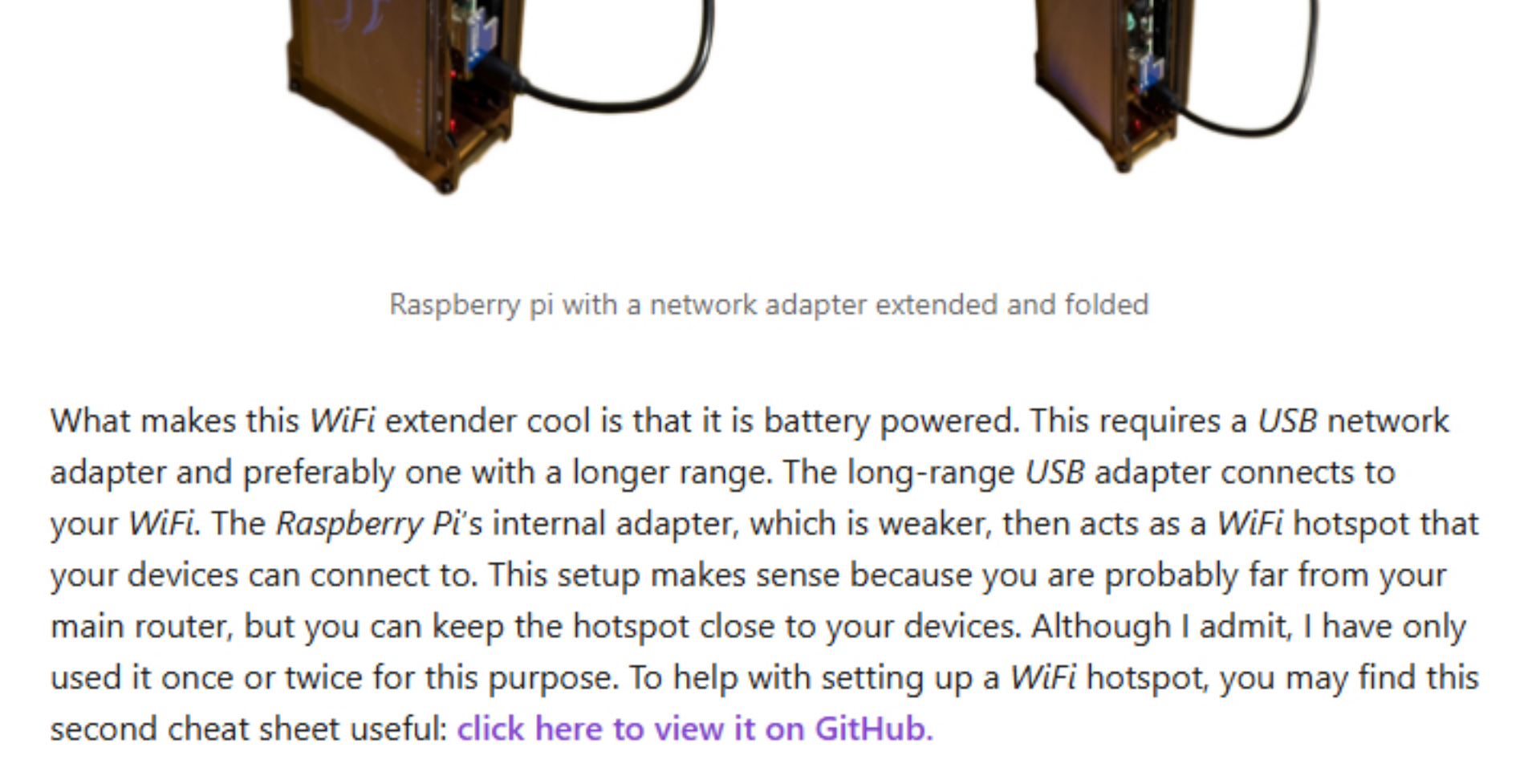


Raspberry pi with a network adapter extended and folded

What makes this *WiFi* extender cool is that it is battery powered. This requires a *USB* network adapter and preferably one with a longer range. The long-range *USB* adapters connect to your *WiFi*. The *Raspberry Pi*'s internal adapter, which is weaker, then acts as a *WiFi* hotspot that your devices can connect to. This setup makes sense because you are probably far from your main router, but you can keep the hotspot close to your devices. Although I admit, I have only used it once or twice for this purpose. To help with setting up a *WiFi* hotspot, you may find this second cheat sheet useful: [click here to view it on GitHub](#).

Part 4: Media Device

Due to time constraints, I had to pause my *Raspberry Pi* adventure. Now basically a TV media device. Since it is a computer, having it plugged into the TV is handy. Running *Pi OS* with a remote keyboard and mouse pad, you can open a web browser and go on *YouTube*, *Netflix*, etc. Its size is comparable to a *Roku*, so it does not look too crazy by the TV. Although, in the photo below, you can see that there is a bit of wiring mayhem because of the sound splitter switch, *USB*, *HDMI* and power cables. The *Raspberry Pi* on the bottom right and its beefy network adapter sitting on the speaker.



To wrap this up: the features above are just what I needed to solve specific problems along the way. This is a tiny *Linux* machine, but you can do far more with it than what I covered in this blog. If you're interested in *Linux* but have never used it, this is a great way to get familiar with it. The entry barrier is low, the build is inexpensive compared to a full-size desktop, and there's plenty of documentation and tutorials available.